

AY25 PROGRAM ASSESSMENT REPORT

v2026.1

Tracking Data

CIM code (e.g., ME-CM)	MAS-AI
Program Name	Master of Artificial Intelligence
Assessment Year	2025
Program Assessment Coordinator	Mustafa Bilgic
Year of Current Assessment Plan	1
No Assessment This Year per Plan	☐
LO(s) Assessed	LO#1: Develop and test supervised machine learning algorithms.
Presented at program authority meeting date: <i>Please attach minutes</i>	

Methodology

We assessed “LO#1: Develop and test supervised machine learning algorithms” this year. The goals for LO#1 are listed in Table 1.

Table 1. Goals for Learning Objective #1: Develop and test supervised machine learning algorithms.

Goal 1	Conceptual understanding: Students can distinguish between supervised, unsupervised, and reinforcement learning, and can identify and differentiate between classification, regression, and clustering tasks based on problem characteristics and data structure.
Goal 2	Algorithmic proficiency: Students can demonstrate a mathematical understanding of key supervised learning algorithms (e.g., linear regression, decision trees, support vector machines, neural networks), including their objective functions, optimization techniques, and assumptions.
Goal 3	Data Preprocessing and Feature Engineering: Students can prepare datasets for supervised learning by handling missing data, encoding categorical variables, and standardizing numerical variables.
Goal 4	Model evaluation: Students can evaluate model performance using appropriate metrics (e.g., accuracy, precision, recall, F1, AUC) and validation techniques (e.g., cross-validation).
Goal 5	Hyper-parameter tuning: Students can apply hyper-parameter tuning methods (e.g., grid-search) for model tuning and selection.
Goal 6	Application to real-world problems: Students can apply supervised learning to a real-world dataset and communicate findings effectively.

LO#1 is mapped to “CS 584 Machine Learning.” CS 584 sections 01, 02, and 03 were taught in Spring 2025 by a full-time faculty, Dr. Shouvik Roy. CS 584 sections 04 and 05 were taught by an adjunct. MAS-AI students could take any of these sections. Here are the number of MAS-AI students in each section:

Table 2 Number of MAS-AI students in each section of CS 584 in Spring 2025.

Section	Instructor	# of MAS-AI students
CS584-01	Shouvik Roy	11
CS584-02	Shouvik Roy	4
CS584-03	Shouvik Roy	4
CS584-04	Adjunct	4
CS584-05	Adjunct	1
Total		24

We reached out to the adjunct in Fall 2025 but did not hear back. Hence, this assessment is based on MAS-AI students who took CS 584 sections 01, 02, and 03.

Each of the six goals (listed in Table 1) were mapped to questions in the assignments, the project, and the exams. The mapping and the weight of each question in the respective assignments, project, and exams are listed in Table 3.

Table 3. Mapping of the goals to the assignment, project, and exam questions, along with the weight of each question.

Goal	Mapping
Conceptual understanding: Students can distinguish between supervised, unsupervised, and reinforcement learning, and can identify and differentiate between classification, regression, and clustering tasks based on problem characteristics and data structure.	Assignment 1 - question 1; 70% of the grade. Assignment 2 - questions 1, 2; 78% of the grade. Assignment 3 - questions 3, 4; 50% of the grade. Midterm - questions (long-form) 1, 2, 3, 4; 50% of the grade. Final - questions (long-form) 2, 4, 5; 30% of the grade. Project - 30% of the grade.

Algorithmic proficiency: Students can demonstrate a mathematical understanding of key supervised learning algorithms (e.g., linear regression, decision trees, support vector machines, neural networks), including their objective functions, optimization techniques, and assumptions.	Assignment 1 - questions 1.e, 2.a ; 32% of the grade. Assignment 2 - questions 1.c, 2.b, 2.d, 3.b; 37% of the grade. Assignment 3 - questions 3.a.3, 4.a.3; 20% of the grade. Midterm - questions (short-form) 2, 4, 5, 8, 9, 13.a,b,e,g,j, (long-form) 1, 2, 3, 4; 70% of the grade. Final - questions (short-form) 18, 20, (long-form) 1, 2, 3, 4, 5; 54% of the grade. Project - 50% of the grade
Data Preprocessing and Feature Engineering: Students can prepare datasets for supervised learning by handling missing data, encoding categorical variables, and standardizing numerical variables.	Assignment 1 - questions 1.a, 1.b, 1.c, 1.d; 48% of the grade. Assignment 2 - questions 1.a, 1.b, 2.a, 3.a; 18% of the grade. Assignment 3 - questions 3.a.1, 3.a.2, 4.a.1, 4.a.2; 10% of the grade. Midterm - questions (long-form) 2, 3; 20% of the grade. Final - questions (long-form) 4, 5; 20% of the grade. Project - 20% of the grade
Model evaluation: Students can evaluate model performance using appropriate metrics (e.g., accuracy, precision, recall, F1, AUC) and validation techniques (e.g., cross-validation).	Assignment 1 - questions 1.e, 2.b; 20% of the grade. Assignment 2 - questions 1.d, 1.g, 2.c, 2.d; 25% of the grade. Assignment 3 - questions 3.a.3, 4.a.3; 20% of the grade. Midterm - questions (short-form) 6, 11, 12, 13.h, (long-form) 4; 32% of the grade. Final - questions (short-form) 2, 7, (T/F)10; 5% of the grade. Project - 30% of the grade.
Hyper-parameter tuning: Students can apply hyper-parameter tuning methods (e.g., grid-search) for model tuning and selection.	Assignment 1 - question 1.e; 12% of the grade. Assignment 2 - questions 1.c, 2.d, 3.b; 23% of the grade. Assignment 3 - questions 3.b, 4.b; 10% of the grade. Midterm - questions (short-form) 3, 13.f,i; 6% of the grade. Final - questions (short-form) 6, (T/F)4; 6% of the grade. Project - 20% of the grade.
Application to real-world problems: Students can apply supervised learning to a real-world dataset and communicate findings effectively.	Assignment 1 - questions 1.e, 1.f, 2.a, 2.b ; 52% of the grade. Assignment 2 -questions 1.e, 1.f, 3.b, 3.c, 3.d; 33% of the grade. Assignment 3 - questions 3.c, 4.c; 10% of the grade. Midterm - questions (long-form) 2, 3, 4; 40% of the grade. Final - questions (long-form) 4, 5; 20% of the grade. Project - 100% of the grade.

The assignments, the project description, the midterm exam, and the final exam are attached as a supplement. Additionally, four artifacts (two sample midterm exam submissions, and two final exam submissions) are attached as a supplement.

Results

Data tables

The score of each student on each assignment, the project, the midterm exam, and the final exam are presented in Table 4.

Table 4. Assignment, project, and exam scores of each student. The student names are anonymized.

Name	Assignment 1 (50)	Assignment 2 (70)	Assignment 3 (100)	Project (150)	Midterm exam (50)	Final exam (100)
Student 1	50	63	82	141	33	77
Student 2	50	67	93	148	46	96
Student 3	50	66	93	149	38	89
Student 4	48	65	93	141	39	77
Student 5	50	70	91	142	38	94
Student 6	50	69	89	146	39	79
Student 7	50	65	98	148	45	95
Student 8	50	67	87	147	38	78
Student 9	47	68	94	146	33	91
Student 10	48	65	84	145	40	89
Student 11	50	68	98	150	49	66
Student 12	50	70	100	148	38	84
Student 13	50	67	96	145	36	57
Student 14	50	65	98	142	47	83
Student 15	50	68	100	144	45	80
Student 16	50	62	100	144	47	93

Student 17	50	55	100	150	45	63
Student 18	46	67	88	144	40	67
Student 19	50	68	98	146	40	62

For each goal (listed in Table 1), the percentage scores for each student are presented in Table 5.

Table 5. Percentage scores for each student for each goal.

Student	Goal 1	Goal 2	Goal 3	Goal 4	Goal 5	Goal 6
Student 1	78%	76%	80%	78%	84%	82%
Student 2	95%	95%	96%	95%	96%	96%
Student 3	87%	86%	88%	86%	91%	90%
Student 4	83%	81%	83%	84%	87%	86%
Student 5	88%	87%	90%	86%	92%	89%
Student 6	85%	83%	85%	86%	89%	88%
Student 7	94%	94%	95%	94%	95%	95%
Student 8	84%	81%	84%	84%	88%	87%
Student 9	84%	83%	85%	81%	89%	86%
Student 10	87%	87%	88%	87%	90%	90%
Student 11	89%	86%	86%	95%	91%	92%
Student 12	87%	84%	87%	86%	91%	89%
Student 13	76%	72%	75%	80%	82%	81%
Student 14	92%	90%	91%	93%	92%	92%
Student 15	90%	88%	89%	92%	91%	91%
Student 16	94%	94%	95%	95%	94%	95%
Student 17	84%	81%	82%	90%	86%	89%
Student 18	81%	78%	80%	84%	85%	85%
Student 19	81%	77%	79%	85%	85%	85%

Table 6. Whether the students Does not Meet (D), Meets (M), or Proficient (P) in each of the goals. The right-most three columns and the bottom three rows present the counts of each category.

Student	Goal 1	Goal 2	Goal 3	Goal 4	Goal 5	Goal 6	D	M	P
Student 1	D	D	M	D	M	M	3	3	0
Student 2	P	P	P	P	P	P	0	0	6
Student 3	M	M	M	M	P	P	0	4	2
Student 4	M	M	M	M	M	M	0	6	0
Student 5	M	M	P	M	P	M	0	4	2
Student 6	M	M	M	M	M	M	0	6	0
Student 7	P	P	P	P	P	P	0	0	6
Student 8	M	M	M	M	M	M	0	6	0
Student 9	M	M	M	M	M	M	0	6	0
Student 10	M	M	M	M	P	P	0	4	2
Student 11	M	M	M	P	P	P	0	3	3
Student 12	M	M	M	M	P	M	0	5	1
Student 13	D	D	D	M	M	M	3	3	0
Student 14	P	P	P	P	P	P	0	0	6
Student 15	P	M	M	P	P	P	0	2	4
Student 16	P	P	P	P	P	P	0	0	6
Student 17	M	M	M	P	M	M	0	5	1
Student 18	M	D	M	M	M	M	1	5	0
Student 19	M	D	D	M	M	M	2	4	0
D	2	4	2	1	0	0			
M	12	11	12	11	9	11			
P	5	4	5	7	10	8			

Description of findings

The bottom three rows of Table 6 show that most students Meet (M) the goals for LO#1. Additionally, the number of students who are Proficient (P) are higher than the number of students who do not meet the goals (D). The right-most three columns show that often fall into one of D, M, or P categories in each of the goals. That is, some students do well (P) in in all goals and while few students do not meet the requirements (D) for most goals. While most students are doing well, Students 1 and 13 are of particular concern.

Continuous Improvement, CI, plans

Prior CI plan impact

This is the first time we are assessing LO#1 and hence no prior CI plan exists.

☒ no previous CI plans available

This cycle's CI plan

While the faculty who teach the course are aware of the learning objectives for the program, they were not provided the six subgoals (listed in Table 1) ahead of the time. Next time the course is offered, the instructors will be provided with the learning objectives and the goals specific to the MAS-AI program.

It is also important for the instructors to incorporate these learning objectives and goals into their syllabi. Doing so should help students understand the focus and objectives of the course, and not lost in minute details.

The program director will share these learning goals and objectives with the course instructors.